

Chapter - 11

COMPUTERS

11.1 Introduction to Computer

- A **digital computer** is an electronic device that works on the binary number system (base-2), representing values using two digits, 0 and 1 (known as Binary Digits or Bits). All digital computers use binary internally owing to its straightforward implementation in electronic circuitry.
- **Bit** (Binary Digit) is the smallest unit of storage in a computer. **One Byte** consists of 8 bits and typically represents one character.
- Computer data storage capacity is expressed as: **KiloBytes** (KB = 1,024 Bytes), **MegaBytes** (MB), **GigaBytes** (GB), **TeraBytes** (TB = 1,024 GB), and **PetaBytes** (PB = 1,024 TB) — reflecting the massive growth in modern storage needs.
- Modern computers use a high-frequency **clock** (a pulse train of 0s and 1s) to synchronise operations. Clock speed is measured in GHz (GigaHertz). Current multi-core processors in personal computers typically operate at 3–5 GHz per core, with multiple cores enabling parallel execution.
- Computers support **multimedia** data — text, images, audio, video, and interactive content. Form factors now include desktop PCs, laptops, tablets, smartphones, and embedded computing devices.

a) Definition of Computer

A computer is an electronic device that:

- Operates under control of instructions stored in its own memory unit
- Accepts data from various input sources
- Processes data arithmetically and logically at high speed
- Produces output of processing and stores results in permanent or temporary media

b) Types and Classification

- **Desktop / Laptop (Notebook) Personal Computer** – the standard computing device for office and engineering work. Laptops integrate all components into a portable, battery-powered form factor.
- **Tablet / Smartphone** – touch-based mobile computing devices running operating systems like Android or iOS. They are increasingly used for field data entry, monitoring, and approvals in industrial environments.
- **Workstation** – a powerful desktop computer with enhanced CPU, GPU, and memory capabilities, used for CAD/CAM, simulation, 3D modelling, and data analytics.
- **Server** – optimised to provide services to other computers over a network. Types include Database Server, Application Server, Web (HTTP) Server, Mail Server, File Server, and Virtualisation Server.
- **Edge Computing Devices** – compact, ruggedised computers deployed close to the production process (e.g., on PLCs, sensors, and IoT gateways) to process data locally and reduce latency.
- **Mainframe** – high-throughput systems used by large organisations for bulk transaction processing (banking, insurance). Still relevant for mission-critical, high-volume workloads.

- **Supercomputer** – multiple high-performance processors working in parallel. India's **PARAM Siddhi-AI** (built under the National Supercomputing Mission) is among the fastest in Asia. Supercomputers are used for weather forecasting, material science simulation, and AI research.

c) Computer Characteristics and Advantages

- The main characteristics of a computer are **speed, accuracy, versatility, automation**, and the ability to handle repetitive tasks without fatigue.
- Computer knowledge is now essential across all job roles. Digital literacy enables employees to stay current in a rapidly changing technological environment.
- Using computers, one can perform in-depth analysis of operational data and make timely, informed decisions. Predictive analytics can flag process deviations before they escalate.
- **Email, instant messaging, and video conferencing** have replaced paper-based communication, providing real-time connectivity with business partners and colleagues across the globe.
- Computers provide highly accurate calculations and reports. Computerised financial statements and technical reports are dependable and consistent regardless of who prepares them.
- **Automation** of repetitive tasks through computers results in higher productivity, reduced errors, lower operating costs, and measurable benefit to the organisation.

d) Computer Generations

- **First Generation (1940s–50s):** Used thousands of vacuum tubes (thermionic valves). Example: ENIAC (1945, USA) — the first general-purpose electronic computer. These machines were large, power-hungry, and unreliable.
- **Second Generation (late 1950s–60s):** Vacuum tubes replaced by **transistors**, making computers smaller, faster, cheaper, and more reliable. Assembly language and early high-level languages emerged.
- **Third Generation (1960s–70s):** Introduction of **Integrated Circuits (ICs / microchips)** dramatically increased computing power while reducing size and cost. Operating systems and time-sharing emerged.
- **Fourth Generation (1970s–present):** Invention of the **microprocessor** (Intel, 1971) placed an entire CPU on a single chip. Led to personal computers, laptops, and dominance of the PC in homes and offices.
- **Fifth Generation (emerging):** Characterised by **AI-optimised processors** (GPUs, NPUs, TPUs), quantum computing research, and neuromorphic chips. These systems are designed to support machine learning, natural language processing, and autonomous decision-making.

e) Hierarchical System of Computers in Steel Industry (Level I to IV)

A typical manufacturing digital control system has the following levels:

- **Level I** – Programmable Logic Controllers (PLC), Distributed Control Systems (DCS), and Instrumentation/SCADA Data Acquisition systems achieve **direct process control** (e.g., valve actuation, drive control, sensor reading).
- **Level II** – Supervisory monitoring and **process control optimisation**. Software models optimise process parameters in real time (e.g., Blast Furnace burden model, BOF oxygen blowing model). Includes Digital Twin applications.

- **Level III** – Plant-level systems for **production planning, MES (Manufacturing Execution Systems)**, maintenance management, and quality management. Interfaces with both Level II and Level IV.
- **Level IV** – Corporate systems for **business and financial planning**. Example: SAP S/4HANA ERP. Manages finance, HR, procurement, sales, and strategic reporting across the enterprise.
- Communication exists between computers at the same level and between adjacent levels. Modern architectures also integrate cloud platforms for advanced analytics and reporting accessible beyond plant boundaries.

11.2 Hardware and Software Concepts

Understanding Hardware and Software

- **Computer Hardware** includes all physical components: input devices, CPU, memory, output devices, and storage. Hardware is tangible — it can be seen and touched.
- **Computer Software** is a set of programs containing detailed instructions that tell the computer exactly what to do. Software is categorised as **system software** (OS, drivers, utilities) and **application software** (MS Office, ERP, browsers, process control applications).

Parts of Computer and Functions

Central Processing Unit (CPU)

- The CPU executes program instructions. Modern CPUs use **multi-core architecture** (dual-core, quad-core, hexa-core, octa-core and above), allowing parallel execution of multiple tasks simultaneously.
- Instruction types include: input/output, arithmetic, logic, branch, and character manipulation. CPU performance is determined by **core count, clock speed (GHz), and cache size**. ARM-based processors (used in mobile devices and increasingly in servers) offer high performance per watt.

Main Memory

- **Read Only Memory (ROM)** stores firmware permanently. Contents are retained even when power is off (non-volatile). Modern systems use **UEFI firmware** stored in flash-based ROM chips, replacing the older BIOS.

Types of ROM:

- o **Mask-Programmed ROM (MPROM)** – programmed at the factory; cannot be changed.
- o **PROM (Programmable ROM)** – user-programmable once using special hardware.
- o **EPROM** – erasable using ultraviolet light and reprogrammable.
- o **EEPROM / Flash ROM** – electrically erasable and reprogrammable in-system. Used in pen drives, SSDs, and BIOS/UEFI chips.

- **Random Access Memory (RAM)** is volatile working memory — data is lost when power is removed. It holds the operating system, active applications, and data being processed. Modern PCs typically have **8 GB to 64 GB** of RAM.

Types of RAM:

- o **SRAM (Static RAM)** – retains data as long as power is supplied; very fast; used in CPU cache memory.
- o **DRAM (Dynamic RAM)** – requires periodic refresh; used as main memory; slower than SRAM but higher density.
- o **DDR4 SDRAM** – current mainstream standard, running at speeds of 2133–3200 MT/s. Transfers data on both rising and falling clock edges.
- o **DDR5 SDRAM** – latest standard, offering double the bandwidth of DDR4 (up to 6400 MT/s), lower power, and higher capacity per module — now standard in new-generation servers and workstations.

Secondary Storage

- **Solid State Drives (SSDs)** are the primary storage medium in modern computers. Using NAND flash memory with no moving parts, SSDs offer much faster read/write speeds (500 MB/s to over 7,000 MB/s for NVMe SSDs), lower power consumption, and higher durability compared to HDDs.
- **Hard Disk Drives (HDDs)** use magnetic platters on a rotating spindle with read/write heads on circular tracks. Capacities now range from 1 TB to 20+ TB. HDDs remain cost-effective for high-capacity archival and backup storage.
- **Network Attached Storage (NAS) / Storage Area Networks (SAN)** provide centralised, high-availability storage accessible over a network — standard in enterprise data centres for shared file storage and database volumes.
- **Cloud Storage** (e.g., Microsoft Azure Blob Storage, AWS S3) stores data on remote servers accessible via the internet or private WAN. It provides scalable, geo-redundant storage for backups, archives, and collaborative document management.
- **Pen Drives (USB Flash Drives)** use flash memory integrated with a USB connector. Modern variants support **USB 3.1/3.2** and **USB-C** for high-speed transfers. They remain widely used for portable data transfer.
- **Optical media** (CD/DVD) are largely superseded for routine use but may still be encountered for legacy data or software distribution.

Input Devices

- Input devices feed data into the computer. Common examples: **keyboard, mouse, touchpad, touchscreen, barcode/QR scanner, biometric reader** (fingerprint, face recognition), **webcam, microphone**, and **digitiser/graphics tablet**.
- In industrial environments, **RFID readers, barcode scanners, ruggedised handheld terminals**, and **IoT sensors** serve as automated input devices, feeding process data directly into plant computer systems without manual entry.
- **Voice input** through digital assistants and speech-to-text engines is increasingly used for hands-free data entry in shop-floor and field applications.

Output Devices

- Output devices present processed data to the user. Examples: **monitor, printer, speaker, projector, plotter, and industrial display panels**.
- Modern monitors use **LED-backlit LCD** or **OLED** technology. They offer higher resolution (Full HD 1920×1080, 4K 3840×2160), wider colour gamut, and lower power consumption than older CRT and early TFT displays.

- Printer types include **laser** (fast, economical for high-volume monochrome/colour), **inkjet** (high-quality colour output), and **multifunction printers** (print/scan/copy/email). Network printers allow shared access across departments.
- A **device driver** is software enabling the operating system to communicate with a hardware peripheral. Modern operating systems (Windows 11, Linux) automatically detect and install drivers for most standard devices via online repositories.

Digital Computer Data Representation

- **Integer data** (whole numbers) is stored in 4-byte (32-bit) or 8-byte (64-bit) blocks. 64-bit computing, now universal, supports far larger numbers and memory addressing than 32-bit systems.
- **Floating-point (real) numbers** are stored in 4-byte (single precision) or 8-byte (double precision) blocks split into mantissa and exponent, following the IEEE 754 standard.
- **Character data** is stored using **ASCII** for basic English characters and **Unicode (UTF-8/UTF-16)** for multilingual text — enabling support for Hindi, Tamil, Chinese, Arabic, and all world scripts in a single unified encoding standard.

11.3 Applications of Computers in Steel Industry

Some of the key computerised functions are listed below:

Finance & Accounts

- Bills and claims processing, stores and sales accounting
- Stock and asset management, payroll processing, budgeting and control
- Real-time financial dashboards and management reporting via ERP

Materials Management

- Item Master and Vendor Master management; e-tendering and online procurement
- Receipt, storage, issue, inventory tracking with barcode/RFID integration
- Supplier Relationship Management (SRM) for vendor performance evaluation

Human Resource

- Employee Master, reporting relationships, nomination management
- Recruitment, promotion, transfer, separation, and performance management systems
- Self-service HR portals enabling employees to access payslips, apply for leave, and update information online

Process Control Applications

- Optimisation of coke oven operations for minimum energy use, maximum coke production, and quality control
- Moisture, charge level, composition, and fuel optimisation for Sinter Plant operations
- Heat and mass balance optimisation, stove scheduling, and burden distribution models for Blast Furnace
- Prediction of oxygen blowing profiles, flux additions, and slag composition in BOF steelmaking
- Secondary cooling control, cut-length optimisation, and slab tracking in continuous caster
- Reheating furnace control, mill setup calculations, gauge control, width control, and in-process material tracking in Rolling Mills

Industry 4.0 and IoT Applications

- **Condition Monitoring Systems (CMS)** using vibration sensors, thermal cameras, and acoustic sensors for predictive maintenance of critical equipment — reducing unplanned downtime.
- **IoT-connected sensors** across production units feed real-time data to centralised platforms, enabling remote monitoring, anomaly detection, and automated alerts.
- **Artificial Intelligence / Machine Learning** models for quality prediction, yield optimisation, energy benchmarking, and process anomaly detection are increasingly deployed at Level-2 and Level-3.
- **Unified Plant View (UPV)** integrates data from DCS, PLC, Level-1, and Level-2 systems into a single web-based dashboard accessible across the plant and via mobile devices.

Application Integration using ERP

- **Enterprise Resource Planning (ERP)** is a software system based on open industry standards, centrally deployed and globally accessed, designed to eliminate data duplication and integrate all business functions.
- ERP addresses all information needs of an enterprise with a process view, integrating finance, HR, procurement, production, sales, quality, and maintenance into a unified platform.
- **SAP S/4HANA** is the current-generation ERP adopted by SAIL, running on an in-memory database (HANA) for real-time analytics and transaction processing with significantly faster performance than earlier SAP ECC 6.0.
- Along with ERP, **Business Process Re-engineering** eliminates non-value-adding processes and automates workflows. Benefits include best-practice solutions, enterprise-wide information sharing, better supplier/customer interactions, and improved profitability.

11.4 Operating Systems and Computer Architecture

- An **Operating System (OS)** manages hardware resources and provides an interface for users and applications. It performs memory management, process scheduling, file management, device I/O, and networking.
- Most operating systems use **pre-emptive multitasking**, allocating CPU time among concurrent processes based on priority. Modern OS kernels support hundreds of concurrent threads across multiple cores.
- **Common Operating Systems:** Microsoft **Windows 11** (dominant in enterprise desktops), **Linux** (various distributions — RHEL, Ubuntu, CentOS — used extensively on servers and industrial systems), **macOS** (Apple workstations), and **Android/iOS** (mobile and tablet devices).

Host-Centric, 2-Tier, 3-Tier, and Cloud Architectures

- **Host-Centric (Legacy):** Programs stored on a central server with dumb terminals for operator interface. Example: SUN-Solaris server with Oracle RDBMS and text-based forms. These systems are now largely replaced but may still exist in older plants.
- **2-Tier (Client-Server):** Server and client PCs interact over LAN. The client runs the application logic locally. Example: SCADA systems where operator workstations directly query a process database server.

- **3-Tier (Web-Based):** Application Server, Database Server, and Client (web browser) are separate tiers. The browser accesses applications via URL. Most modern plant-level portals, dashboards, and ERP thin clients use this model.
- **Cloud / Hybrid Architecture:** Applications and data are hosted in cloud data centres (Microsoft Azure, AWS, Google Cloud) and accessed over secure internet or WAN links. Hybrid cloud combines on-premise plant systems with cloud services for analytics, AI, and disaster recovery. This is increasingly adopted for ERP, collaboration tools, and data lakes.

Online, Offline, and Real-Time Processing

- **Batch (Offline) Jobs:** Pre-configured jobs running to completion without human interaction, with all input pre-selected. Example: payslip preparation, monthly MIS report generation, data archival jobs.
- **Online (Interactive) Jobs:** Prompt the user for input in real time. Examples: financial transactions, indenting systems, railway reservation, leave applications.
- **Real-Time Processing:** System correctness depends not only on logical accuracy but also on **timing** — results must be produced within defined time constraints. Examples: optimised cut-length set point download in the caster, dynamic reheating furnace set points, autonomous mill gauge control.
- Industrial systems in steel plants use a combination of all three processing types, often integrated through MES and Level-2 automation systems.

Boot Procedure and Shutdown

- **Bootstrapping (Cold Boot):** When a computer is powered on, it activates firmware (UEFI) from ROM. Steps: (1) POST (Power-On Self Test), (2) identify boot device and load OS bootloader, (3) initialise hardware, file system check, network interfaces, and (4) present the OS login screen.
- **Proper Shutdown:** A computer must always be shut down using the OS shutdown command. This stops running processes, flushes file system buffers, and writes pending data to disk cleanly. Abrupt power-off risks data corruption and file system inconsistency.

11.5 Computer Language and Application Software

Programming Concepts

- To solve a problem, an **algorithm** — a finite sequence of precise instructions — must first be designed. Algorithms are often expressed graphically as **flowcharts** or as **pseudocode** before being coded in a programming language.
- A **high-level programming language** is a human-readable notation used to express algorithms. The compiler or interpreter translates high-level code into machine-executable binary instructions.

Program Development Lifecycle

- Analyse user requirements thoroughly before design begins
- Design the solution architecture, data model, and software framework
- Develop database structures and program code with version control (e.g., Git)
- Test for quality — unit testing, integration testing, and user acceptance testing (UAT)
- Deploy the solution; conduct post-implementation review and corrections
- Continuous improvement, performance monitoring, and supportive maintenance

Examples of High-Level Computer Languages

- **Python** — now the most widely used language for data analytics, machine learning, scientific computing, and automation scripts. Extensively used in AI/ML and Industry 4.0 applications.
- **C / C++** — used for system programming, embedded systems, real-time process control (Level-1/Level-2 automation), and performance-critical applications.
- **Java / C#** — used for enterprise application development, ERP modules, and web application backends. Java is platform-independent; C# is prevalent in Microsoft technology stacks.
- **FORTRAN** — legacy language still used in numerical/scientific computation modules in steel process models.
- **JavaScript / TypeScript** — dominant languages for web front-end development and increasingly for server-side (Node.js) applications. Used in plant dashboards and web portals.
- **SQL (Structured Query Language)** — the universal language for relational database operations: create table, select, insert, update, delete. Used across all database platforms (Oracle, SQL Server, MySQL, PostgreSQL).
- Modern development uses **DevOps practices** — combining development and operations with automated testing, continuous integration/continuous deployment (CI/CD) pipelines, and containerisation (Docker, Kubernetes) for faster and more reliable software delivery.

11.6 Data Centre Management

Data Storage in Central Server

- A **Data Centre** is a secured, air-conditioned facility housing server computers, storage systems, and network equipment. It operates 24×7 with uninterruptible power supply (UPS), diesel generators, precision cooling, fire suppression, and physical access control.
- Modern data centres use **rack-mounted servers** with high-capacity SSD and NAS/SAN storage arrays, offering fast random access. **Virtualisation** (VMware, Hyper-V) enables multiple virtual servers to run on shared physical hardware, improving utilisation and flexibility.
- **Cloud Data Centres** (Microsoft Azure, AWS, Google Cloud) offer data centre services on a subscription basis, eliminating the need to maintain all on-premise hardware. Many organisations use a **hybrid model** — keeping sensitive plant data on-premise while using cloud for analytics, DR, and collaboration.
- The role of Data Centre staff is to monitor system health, troubleshoot problems, manage backups, enforce security policies, and coordinate with vendors for hardware and software support.

Backup Devices and Procedures

- Data falls into **transaction data** (day-to-day records) and **master data** (reference records). Regular backups are essential to minimise data loss in case of hardware failure, ransomware attack, or other disasters.
- Modern backup strategies use **disk-based backup** (fast restore) combined with **cloud backup** (offsite, geo-redundant). The **3-2-1 rule** is widely followed: 3 copies of data, on 2 different media types, with 1 copy offsite.
- **Backup schedules** combine full backups (complete copy of all data, typically weekly) with incremental backups (only changes since last backup, taken daily or

more frequently). This minimises backup time and storage use while enabling point-in-time recovery.

- Legacy **magnetic tape and DAT (Digital Audio Tape)** media were previously standard for sequential backup. Modern systems have largely transitioned to **LTO (Linear Tape-Open) tape** for long-term archival (up to 45 TB per cartridge compressed) or cloud archival services.

Data Restore Procedures for Recovery

- In case of server or storage failure, data is restored from the latest full backup plus subsequent incremental backups. Recovery Time Objectives (RTO) and Recovery Point Objectives (RPO) define maximum acceptable downtime and data loss.
- **Disaster Recovery (DR) Sites** at separate geographical locations (different seismic zones) maintain near-real-time data replication, enabling failover with minimal data loss and business continuity within a few hours.

11.7 Network and Connectivity

Local Area Network (LAN), Wide Area Network (WAN)

- A **LAN (Local Area Network)** covers a limited geographic area such as an office building or plant section. LANs offer high data transfer rates (1 Gbps to 10 Gbps using modern switches) and are the backbone of plant-level computer communication.
- A **WAN (Wide Area Network)** connects networks across large geographic distances — inter-plant, inter-city, or global. WANs use leased telecommunication lines, MPLS (Multiprotocol Label Switching) links, or internet VPN tunnels.

Cables and Network Equipment

- **Network media types:** Twisted pair cable (**Cat 6 / Cat 6A UTP/STP** for up to 10 Gbps), **Optical Fibre Cable (OFC/OFC)** for high-speed backbone links (up to 100 Gbps) and long-distance runs, and **wireless (Wi-Fi)** for flexible access.
- **Wi-Fi 6 (802.11ax)** is the current standard for enterprise wireless LANs, offering throughput up to 9.6 Gbps, improved performance in dense environments, and lower latency — suitable for plant monitoring applications.
- **5G connectivity** provides high-speed, low-latency wireless connectivity for industrial IoT, autonomous vehicles, and mobile equipment monitoring within large plant areas where cabling is impractical.
- **IP Address (Internet Protocol Address):** A unique numeric address assigned to each network device. **IPv4** uses 32-bit addresses (e.g., 192.168.1.10). **IPv6** uses 128-bit addresses (e.g., 2001:db8::1) and is increasingly adopted to address the exhaustion of IPv4 addresses. The `ipconfig` (Windows) or `ip addr` (Linux) command displays a device's IP address.

Network Devices

- **Router** – forwards packets between different networks (different IP address ranges). Connects the plant LAN to the WAN or internet. Modern routers include firewall, VPN, and Quality of Service (QoS) functions.
- **Network Switch** – connects multiple devices within a LAN segment. Managed switches (supporting VLANs, port security, and traffic monitoring) are used in industrial networks for security and reliability.

- **Firewall** – a security device (hardware or software) that monitors and controls incoming and outgoing network traffic based on defined security rules. Essential for protecting plant networks from cyber threats.
- **Modem / OFC Converter** – interfaces between different transmission media. Optical-to-electrical converters are used where fibre backbone connects to copper Ethernet equipment.

Ports for Network Connectivity

- **Ethernet Port (RJ-45):** Standard network interface on all modern PCs and servers, supporting 1 Gbps or 10 Gbps. Connected to network switches via Cat 6 cable.
- **USB Ports (USB 3.0/3.1/USB-C):** Universal connectivity for peripherals — printers, keyboards, mice, cameras, external drives. USB-C supports data, video, and power over a single connector.
- **Wi-Fi / Bluetooth:** Wireless connectivity built into modern laptops and desktops. Bluetooth connects short-range peripherals (headsets, keyboards); Wi-Fi connects to the network infrastructure.
- **Legacy ports** (serial 9-pin RS-232, parallel 25-pin) are largely obsolete on modern PCs but may still be found on older industrial equipment such as weigh-bridge controllers and legacy PLCs. USB-to-serial adapters are used to interface these.

11.8 Introduction to Windows

Starting Windows and Desktop Navigation

- When the PC is powered on, the **UEFI firmware** performs POST, then loads the **Windows 11** operating system. After entering username and password (or using Windows Hello biometric login), the **GUI desktop** is presented with the Start Menu, Taskbar, and desktop icons.
- **Windows 11** introduces a centred Taskbar, improved virtual desktops (Task View), Snap Layouts for multi-window management, and deep integration with **Microsoft Teams** and **OneDrive** for collaboration and cloud file access.
- Multiple applications (Word, Excel, browser, email) can run simultaneously. Each appears in the Taskbar. **Alt+Tab** switches between open windows; **Windows key + Tab** opens Task View for virtual desktop management.

Maximise, Minimise, Exit Windows

- Each application window has **Minimise, Maximise/Restore, and Close** buttons (top-right). **Snap Layouts** (hover over Maximise button) allows snapping windows side-by-side or in grid arrangements for efficient multitasking.
- Standard menu structure (File, Edit, Insert, Format, View, Help) applies across Microsoft Office applications. The **Ribbon interface** groups commands visually by category.

File Management

- **File Explorer** (Windows key + E) is used to navigate drives, folders, and files. Files can be created, deleted, copied, moved, renamed, and searched. Common file types: `\.docx` (Word), \.xlsx` (Excel), \.pptx` (PowerPoint), \.pdf` (Portable Document), \.jpg/.png` (images).`
- **OneDrive integration** in Windows 11 allows files saved to the OneDrive folder to be automatically synchronised to the cloud and accessible from any device. This serves as a built-in backup and collaboration mechanism.

- Users are responsible for organising their work files in appropriate folders and backing up critical data to approved storage (network drives or OneDrive). File naming conventions and folder structures should follow departmental standards.

Using Pen Drives, Cloud Storage, and Printers

- **Pen drives** are automatically recognised when inserted into a USB port. Files can be copied and deleted using File Explorer. Always use **Safely Remove Hardware** (system tray icon) before physically disconnecting a pen drive to prevent data corruption.
- **Cloud storage (OneDrive, SharePoint)** enables file access and collaboration across devices without physical media. Files can be shared with colleagues and accessed on mobile devices.
- **Printers** are installed via the OS using driver software (downloaded automatically from Windows Update or the manufacturer's website). Network printers are shared across the department — select the printer from the print dialog in any application.

11.9 Office Automation Software (MS Office / Microsoft 365)

MS Word

- A file created in Word is called a **document** (.docx extension). Word is used for note-sheets, inter-office memos, reports, technical documents, and correspondence.
- Key features: margins, page layout, tables, text formatting (font, size, colour, bold, italic), paragraph spacing, bullets and numbering, spell and grammar check, track changes, comments, and mail merge.
- **Microsoft 365 (cloud-based)**: Word documents saved on OneDrive or SharePoint can be co-authored in real time by multiple users simultaneously via a web browser or the desktop app — eliminating version conflicts in collaborative document preparation.

MS Excel

- A file created in Excel is called a **workbook** (.xlsx extension). A workbook contains one or more worksheets. Excel is used for data analysis, financial calculations, MIS reports, and dashboards.
- Data is entered in **cells** addressed by column letter and row number (e.g., A1, B23). Cells accept numbers, text, formulas, and functions. Formula results update automatically when referenced data changes.
- Advanced features include **PivotTables** (for summarising large datasets), **Power Query** (for importing and transforming data from multiple sources), conditional formatting, data validation, VLOOKUP/XLOOKUP, and dynamic charts.
- Chart types available: bar, column, line, pie, scatter (X-Y), area, and combination charts. Charts are linked to live data and update automatically.

MS PowerPoint

- A file created in PowerPoint is a **presentation** (.pptx extension). PowerPoint is used for slide-based presentations in meetings, reviews, training sessions, and management briefings.
- Features: slide layouts, design themes, animations, transitions, embedded videos, charts, and SmartArt. **Presenter View** allows the speaker to see notes and upcoming slides on a separate screen while the audience sees only the presentation.

- Presentations can be shared via **Microsoft Teams** for online meetings, exported as PDF for print distribution, or saved as video for asynchronous viewing — without requiring PowerPoint to be installed on the viewer's device.

11.10 Database Concepts

Data and Information

- **Data** refers to raw numbers, characters, images, or signals that can be stored and processed by a computer. Data has no inherent meaning in isolation. For example: 12345, 10000.0, 1000.0 are raw data values.
- **Information** is data that has been processed and given meaning through context and relationships. For example: Employee No. = 12345, Basic Pay = Rs. 10,000, DA = Rs. 1,000 — this is meaningful information enabling payroll processing and decision-making.

Structured Storage of Data in Rows and Columns

- A **database** is a structured collection of records organised for efficient storage, retrieval, update, and management. A **Database Management System (DBMS)** controls data storage, access, integrity, and security.
- **Relational databases** store data in tables (rows and columns). Tables are linked through common key fields, eliminating data duplication. Examples: Oracle, MS SQL Server, MySQL, PostgreSQL. **SQL (Structured Query Language)** is used to query and manipulate data.
- **NoSQL databases** (e.g., MongoDB, Cassandra, InfluxDB) handle unstructured or semi-structured data — time-series sensor data, JSON documents, log files — increasingly important for IoT and Industry 4.0 data platforms.
- **Big Data platforms** (Apache Hadoop, Apache Spark) are used for processing extremely large datasets from multiple plant systems, enabling advanced analytics, machine learning model training, and historical trend analysis.

11.11 Intranet and Internet

Program Implementation Architectures

- **Host-Centric (Legacy):** Programs on a central server accessed via dumb terminals. Example: SUN-Solaris with Oracle RDBMS. Most such systems have been replaced, but legacy installations may still exist.
- **2-Tier (Client-Server):** Server and client PCs on LAN. Application logic runs partly on the client. Example: older SCADA systems.
- **3-Tier (Web-Based):** Application server, database server, and browser-based client. Example: plant portals accessible via URL. Most modern plant applications use this architecture.
- **Cloud-Based / Microservices:** Applications decomposed into independently deployable services, hosted on cloud infrastructure. Accessed via APIs. Enables scalability, rapid updates, and integration with AI/ML services.

Definitions of Intranet and World Wide Web

- An **Intranet** is a private organisational network using TCP/IP protocols, accessible only to authorised employees. It hosts internal portals, document repositories, ERP thin clients, HR self-service, and departmental homepages. It increases internal communication and reduces paper distribution cost.

- The **Internet** is the global network of networks carrying information services including email, file transfer, video conferencing, and web pages. It underpins modern digital commerce and communication.
- The **World Wide Web (WWW)** is the collection of all publicly accessible websites and web resources on the Internet, accessed through web browsers using HTTP/HTTPS protocols.
- **Cloud Computing** delivers computing services — storage, processing power, databases, analytics, and AI — over the internet. Organisations use cloud for disaster recovery, scalable analytics, ERP hosting, and collaboration tools, reducing on-premise infrastructure costs.

Web Browsers, Search Engines, Email, and Communication Tools

- **Email:** Electronic mail enables composing, sending, storing, and receiving messages. Platforms include Microsoft Outlook (enterprise standard), Gmail, and organisational mail systems. Employees must be vigilant about **phishing emails** — fraudulent messages designed to steal credentials or spread malware. Never open attachments from unrecognised senders.
- **Web Browsers:** Software for accessing and displaying web content. Modern browsers include **Google Chrome**, **Microsoft Edge** (Chromium-based), and **Mozilla Firefox**. Browsers support extensions, integrated PDF viewing, secure HTTPS connections, and developer tools. Internet Explorer is discontinued and no longer supported.
- **Video Conferencing:** Tools like **Microsoft Teams**, **Zoom**, and **Google Meet** enable virtual meetings, screen sharing, file collaboration, and webinars — essential for hybrid and remote working models and inter-plant coordination.
- **Search Engines:** Web-based information retrieval tools. Examples: Google (www.google.co.in), Bing (www.bing.com). Effective searching uses specific keywords, quotation marks for exact phrases, and date filters.
- A **website** is a location on the WWW managed by an individual or organisation, consisting of a home page and linked pages. It is accessed by entering its URL in a browser.

Examples of Important Websites

- SAIL: www.sail.co.in
- Indian Railways information: www.indianrail.gov.in
- Railway e-Ticketing: www.irctc.co.in
- Income Tax e-Filing: www.incometax.gov.in
- Government Services Portal: www.india.gov.in
- Search Engine: www.google.co.in
- The Hindu newspaper: www.thehindu.com

Advantages of the Web

- Establish organisational presence globally, round-the-clock; provide technical support and customer service
- Online advertising, multimedia content delivery, and e-commerce (product catalogues, e-tendering, sales portals)
- **Electronic payment** systems for salary, vendor payments, and procurement
- **Collaboration platforms** enabling real-time document co-authoring, virtual meetings, and project management across locations

11.12 Introduction to ERP

- **ERP (Enterprise Resource Planning)** is an integrated software system that unifies all departments and business functions of an enterprise on a single platform with a shared database, eliminating data silos and enabling real-time cross-functional information sharing.
- ERP promotes seamless integration of all information flows across the organisation, adopting best global business practices. It supports multiple currencies, languages, and regulatory frameworks for multinational operations.
- **SAP S/4HANA** is the current-generation ERP adopted by SAIL, built on the SAP HANA in-memory database platform. It delivers real-time analytics embedded within transactions, eliminating batch reporting delays of older systems. SAP stands for Systems, Applications, and Products in Data Processing.
- ERP implementation requires infrastructure upgrades (servers, network, disaster recovery), process re-engineering, user training, and change management. Benefits include integration, reduced operating costs, faster access to accurate information, and uniform global reporting.

ERP Data Centres and Infrastructure

- Plants with ERP have dedicated **primary and secondary data centres** with servers, firewalls, network switches, storage systems, and physical access controls. Real-time data mirroring between data centres ensures automatic failover with no user disruption if one centre goes down.
- A **Disaster Recovery (DR) site** at a separate geographical location (different seismic zone) provides business continuity with a changeover time of a few hours and minimum data loss. Redundant network routes and diverse fibre links ensure connectivity resilience.
- Network infrastructure is designed with redundant paths, VLAN segmentation, and firewalls to support ERP performance, security, and integration with metal exchanges, SAIL corporate network, and internet interfaces.

ERP Benefits

Tangible Benefits:

- o Integration of all business functions; reduction in operating costs through improved process coordination
- o Improved productivity of processes and personnel; lead time reduction in procurement and production
- o Facilitates day-to-day management with real-time dashboards and drill-down reporting

Intangible Benefits:

- o Organisational transparency and accountability; faster and more accurate access to information
- o Supports strategic planning; uniform reporting according to global standards
- o Accessibility from anywhere via web browser or mobile; supports remote and multi-location management

Important SAP Modules

- **FICO (Financial Accounting & Controlling):** General Ledger, Accounts Receivable/Payable, Asset Accounting, Cost Centre Accounting, Product Costing, and Profitability Analysis.

- **PP (Production Planning):** Annual, monthly, weekly, and daily production planning; capacity scheduling; production order tracking; material consumption booking; shift-end and day-end reporting.
- **PM (Plant Maintenance):** Preventive maintenance planning, breakdown recording, condition-based maintenance, warranty management, and outsourcing of maintenance activities.
- **SD (Sales and Distribution):** Order management, dispatches, pricing, billing, rebates, sales information system, and real estate management.
- **MM (Materials Management):** Procurement, goods receipt, inventory management, vendor evaluation, invoice verification. SRM (Supplier Relationship Management) supports e-tendering and online vendor interaction.
- **QM (Quality Management):** Quality planning, inspection, certification (test certificates), and quality control across the supply chain — from raw materials through finished products.
- **PS (Project System):** Project registration, budget management, schedule control, cost/resource management, project accounting, and reporting.

11.13 Do's and Don'ts — Computer Standard Operating Procedures

Do's

- Ensure **proper earthing** and stable power supply. Check neutral-to-ground voltage periodically (should remain below 5 V). Use a **UPS (Uninterruptible Power Supply)** to protect hardware and allow safe shutdown during power failures.
- **Startup sequence:** Switch on UPS first, then monitor, then the computer.
Shutdown sequence: Close all applications, then CPU, monitor, and finally UPS. During power failure, save work and shut down before UPS battery is depleted (~20 minutes capacity).
- Always use the **Start Menu > Shut Down** option for a clean shutdown. This closes all open files and processes in an orderly manner, preventing data corruption and file system errors.
- **Connect and power on all peripherals** (printer, monitor, scanner) before powering on the computer.
- **Use strong passwords** — minimum 12 characters combining uppercase, lowercase, numbers, and special symbols. Enable **Two-Factor Authentication (2FA)** wherever available. Never share passwords or write them down. Change passwords periodically as per organisational policy.
- **Install and regularly update antivirus / endpoint protection software.** Keep the operating system and all applications patched and updated to address known security vulnerabilities.
- **Lock your workstation** (Windows key + L) whenever leaving your desk — even briefly — to prevent unauthorised access to systems and data.
- **Back up important data** regularly to approved storage (network drive, OneDrive, or pen drive). Follow the 3-2-1 backup rule: 3 copies, 2 media types, 1 offsite.
- Keep hardware clean. Use dust covers when equipment is not in use. Report any hardware fault or suspicious system behaviour to IT helpdesk promptly and maintain a fault log.
- **Log off** the computer when finishing work for the day or leaving for an extended period.

Don'ts

- **Do not open emails or attachments from unknown or suspicious senders.** Phishing attacks are the most common vector for ransomware and data breaches. Verify sender identity and never click suspicious links.
- **Do not download or install unauthorised software.** Only IT-approved software should be installed on official computers. Unauthorised software may contain malware, spyware, or ransomware.
- **Do not connect personal or unknown USB drives** to official computers without IT approval — they may carry viruses or be used to exfiltrate data.
- **Do not share confidential organisational data** via personal email, WhatsApp, social media, or unauthorised cloud storage platforms. Data handling must comply with organisational information security policy.
- Do not eat or drink near computer equipment to prevent accidental damage.
- Do not move powered-on peripherals or disconnect USB devices without safely ejecting them first.
- Do not switch off the UPS while the PC is running. Do not repeatedly switch external devices on and off as a troubleshooting method without consulting IT support.

Computer Viruses and Cybersecurity

- **Computer viruses** are malicious programs designed to spread from one computer to another, corrupting or deleting data, disrupting operations, or providing unauthorised access to attackers.
- **Ransomware** is a particularly dangerous form of malware that encrypts files and demands payment for decryption keys. It has caused major disruptions in industrial and corporate environments worldwide.
- **Phishing emails** mimic legitimate sources (banks, IT departments, management) to trick users into revealing credentials or installing malware. Always verify unexpected requests through a separate communication channel.
- Install, maintain, and regularly update **endpoint protection (antivirus/anti-malware)** software. Scan pen drives before use. Never open email attachments unless the sender is known and the file is expected.
- **Cybersecurity** is a shared responsibility. Every user plays a critical role in protecting the organisation's digital assets by following security policies, reporting incidents, and maintaining good digital hygiene practices.

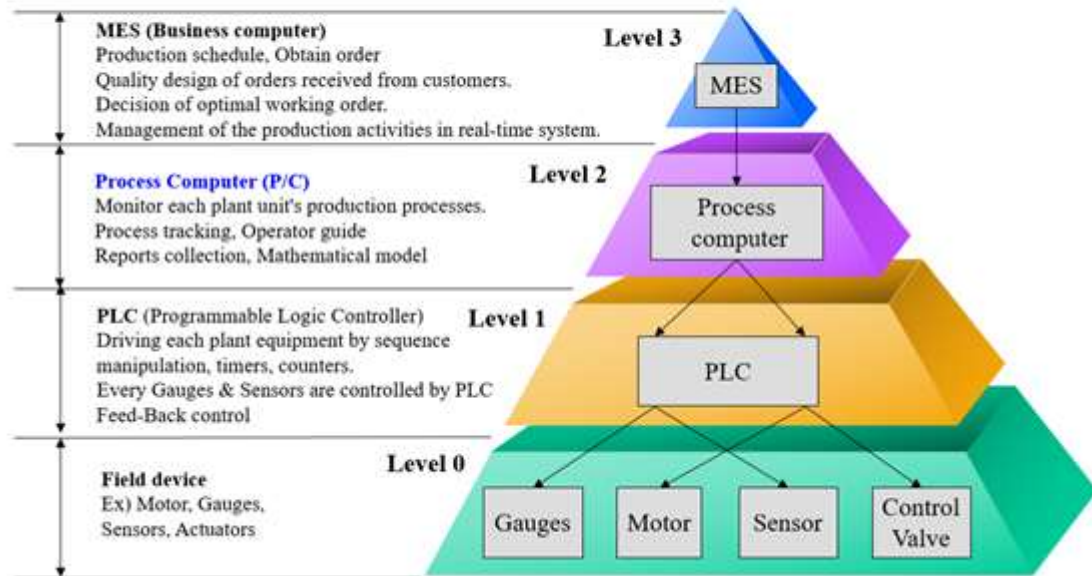
Computer Hardware Maintenance

- Computer maintenance is carried out by the IT/computer department. As a user, raise complaints through the official IT helpdesk with a clear description of the problem. Do not attempt to open or repair hardware yourself. Keep a record of faults reported and actions taken.

11.14 Digital Transformation

Level-2 Automation

Automation System Structure in Industries

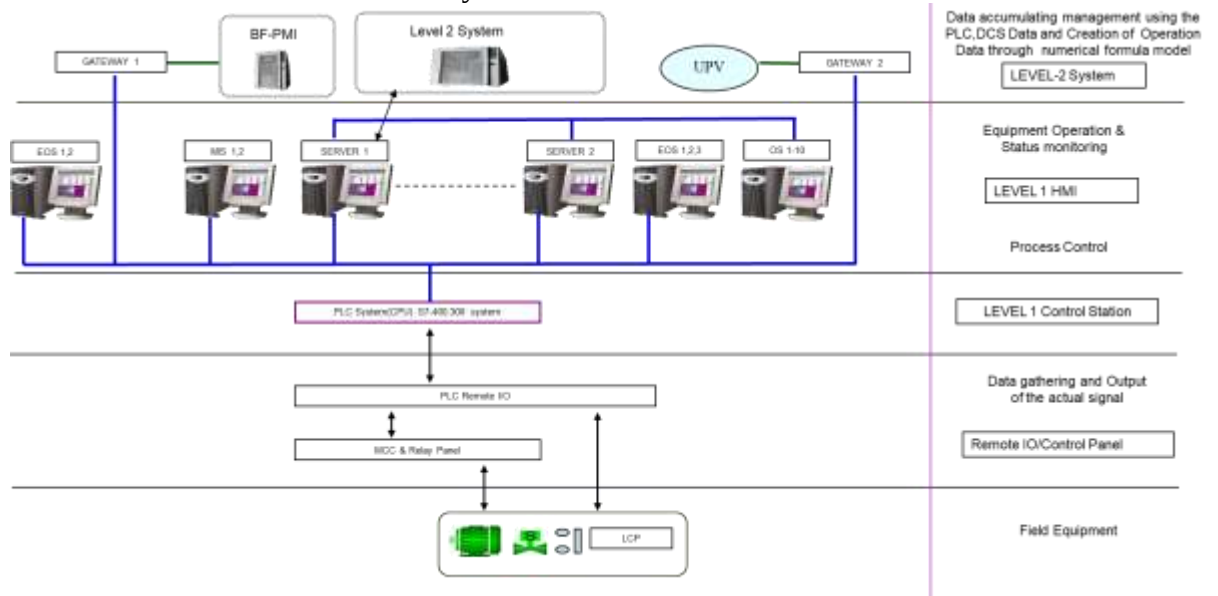


- **Level-2 automation** is used for complex manufacturing processes to optimise operations using **software process models**. The Level-2 system takes all inputs from Level-1 (sensors, actuators) and generates optimised set points for operators and Level-1 controllers.
- **Off-line advisory models** calculate process advice and recommendations for operators based on current conditions, without directly manipulating Level-1 set points.
- **On-line models** directly adjust Level-1 controller set points in real time based on continuous model calculations — enabling closed-loop process optimisation (e.g., BF burden distribution, BOF oxygen blowing, slab reheating furnace curves).
- **Level-2 Systems in Blast Furnace** provide Process Optimisation Models for heat and mass balance, stove scheduling, hearth management, tuyere health, and production forecasting — enhancing operational efficiency, product quality, and safety.
- **Level-2 Systems in SMS / BOF-CCP Complex** perform data exchange between caster and BOF, heat pacing, ferro-alloy consumption modelling, and secondary cooling control — optimising treatment time and alloy consumption.
- Key benefits of Level-2 Automation: uniform operating practice, improved process and product quality, reduction in energy and alloy consumption, optimised treatment time, complete traceability, and operational transparency.
-

Key Components of Level-2 System:

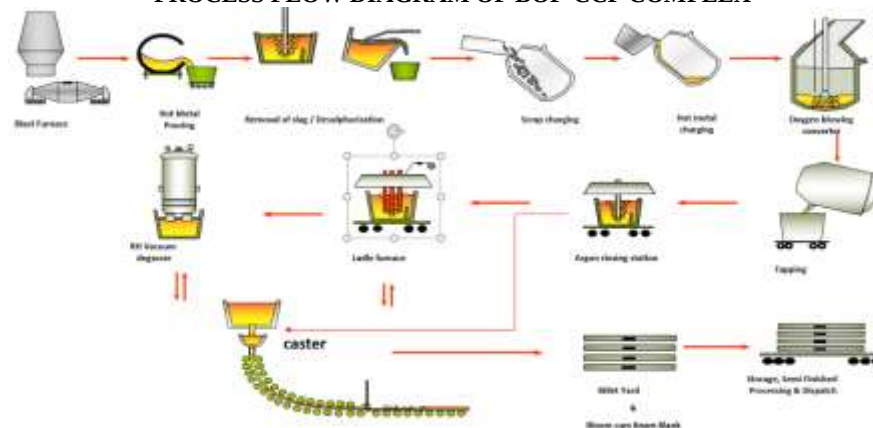


System Architecture:

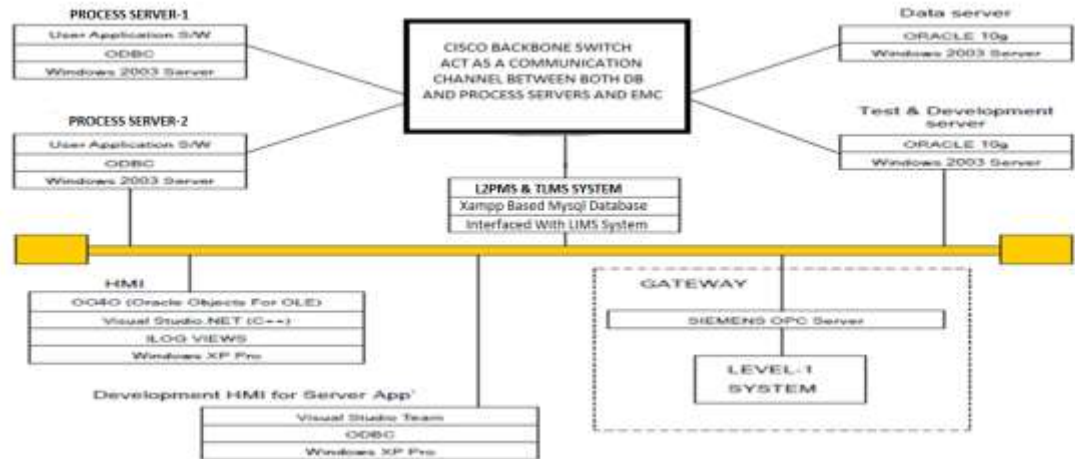


Level 2 Automation Systems of SMS Complex

PROCESS FLOW DIAGRAM OF BOF-CCP COMPLEX



Level 2 Architecture



Industry 4.0

- There have been **four industrial revolutions**: (1) Steam power and mechanisation; (2) Electricity, internal combustion engine, mass production; (3) Digital technology, computers, internet, and automation; (4) **Industry 4.0** — integration of IoT, AI, ML, robotics, cyber-physical systems, and big data in manufacturing.
- Industry 4.0 creates **smart, connected manufacturing ecosystems** where machines, products, and people communicate in real time, enabling autonomous decision-making, predictive maintenance, and mass customisation.

Digitization, Digitalization, and Digital Transformation

- **Digitization** converts analogue information into digital format — e.g., scanning paper drawings into electronic files, converting analogue instrument readings to digital signals.
- **Digitalization** uses digital technologies to transform how businesses operate — adopting mobile devices, cloud software, and automation tools to improve processes and increase efficiency.
- **Digital Transformation** is a holistic rethinking of how the organisation creates value, using digital technologies to create new business models, products, and services, and fundamentally changing interactions with customers, partners, and employees.

Design Principles of Industry 4.0

- **Interoperability**: Machines, devices, sensors, and people connect and communicate seamlessly through standardised industrial protocols (OPC-UA, MQTT, REST APIs).
- **Virtualization**: Digital models (Digital Twins) of physical assets and processes enable simulation and testing before physical implementation, reducing risk.
- **Decentralization**: Decision-making distributed to the lowest appropriate level — edge devices and autonomous controllers — enabling faster response without central system dependency.
- **Real-time data**: Continuous monitoring and optimisation of production processes using live sensor data, enabling faster and more informed decisions.
- **Service orientation**: Value delivered through innovative digital services — predictive maintenance subscriptions, quality analytics platforms, and remote expert support.

Emerging Technologies

- **IoT (Internet of Things):** Connects physical devices and sensors to the internet, enabling collection and analysis of vast amounts of operational data for monitoring, control, and optimisation.
- **AI (Artificial Intelligence) and ML (Machine Learning):** Algorithms that learn from data to identify patterns, make predictions, and support autonomous decision-making — used in quality prediction, anomaly detection, and process optimisation.
- **Robotics and Automation:** Industrial robots and collaborative robots (cobots) automate repetitive, hazardous, or precision-critical tasks — increasing efficiency, consistency, and worker safety.
- **Digital Twin:** A real-time virtual replica of a physical asset or process (e.g., Blast Furnace Digital Twin) that mirrors operational behaviour, enabling remote monitoring, scenario simulation, and predictive analytics.
- **AR (Augmented Reality) and VR (Virtual Reality):** AR overlays digital information on the real world (maintenance guidance, process data displays); VR creates immersive training environments. Both improve decision speed and training effectiveness.
- **Additive Manufacturing (3D Printing):** Creates components by layering materials, enabling rapid prototyping and manufacture of complex spare parts on demand — reducing inventory and lead times.
- **Cybersecurity:** As plants adopt digital technologies, they face new cyber threats. Cybersecurity protects digital assets — systems, networks, and data — from unauthorised access and attack. It requires access controls, encryption, network segmentation, security monitoring (SOC), and a culture of cybersecurity awareness throughout the organisation.

Key Areas Identified For Digital Transformation



High Impact Digital Projects in Steel Plants

Details of High Impact Projects Completed

Development of Online Platforms viz. of Unified Plant View(UPV), Laboratory Information Management System (LIMS), Torpedo Ladle Management System(TLMS) has been done with in-house resources.

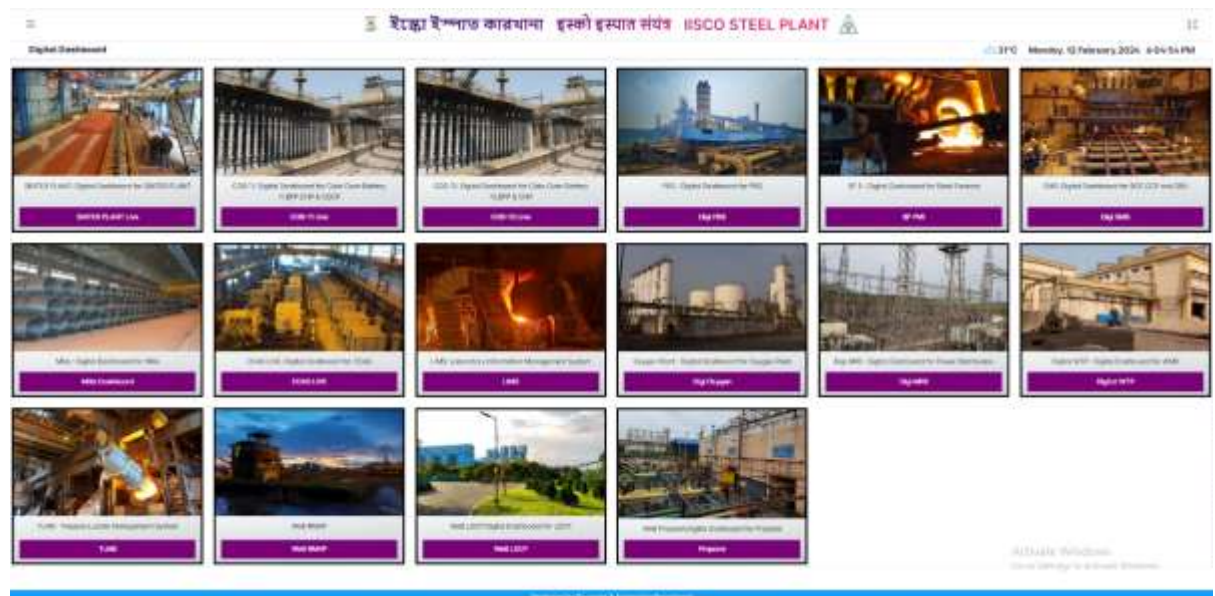
Unified Plant View (UPV) is a single, comprehensive and integrated view of all the plant operations. With Unified Plant View, information from different data silos of various systems and applications are brought together into a single interface, making it easier to manage and monitor your plant operations. The system includes process, production, and techno-economic data of different production and auxiliary units, providing a complete and real-time understanding of the state of the plant.

Unified Plant View is a web based online application which captures real time process data and status of equipment and shops from DCS/PLC/Level 1/Level2 systems. UPV can be accessed from LAN inside the plant and over smart devices (Mobile/Tablet/Laptop) from outside the plant with proper authentication.

Digital Dashboards

Web-based monitoring system of all production and auxiliary units which provides real-time and archived data of critical process parameters for efficient monitoring and management of the entire production process. Operators and managers can quickly identify areas of concern, track key performance indicators, and make data-driven decisions to enhance overall efficiency and performance.

Digital Dashboard for RMHP (WeB- RMHP), Sinter Plant (Sinter-DI), COB-10 & COB-11 (COB LIVE), PBS (Digi-PBS), BF #5 (BF-PMI), LDGP (WeB-LDGP), CCAS (CCAS-Live), SMS (Digi-SMS), Mills (Mills-Online), Oxygen Plant (OxyOnline), WMD (Digital WTP), MRS (Digi MRS) have been developed.



Benefits:

- Real time Monitoring of Process, Production & CMS for Critical Equipment
- Digital view of important process
- Machines operation in real-time

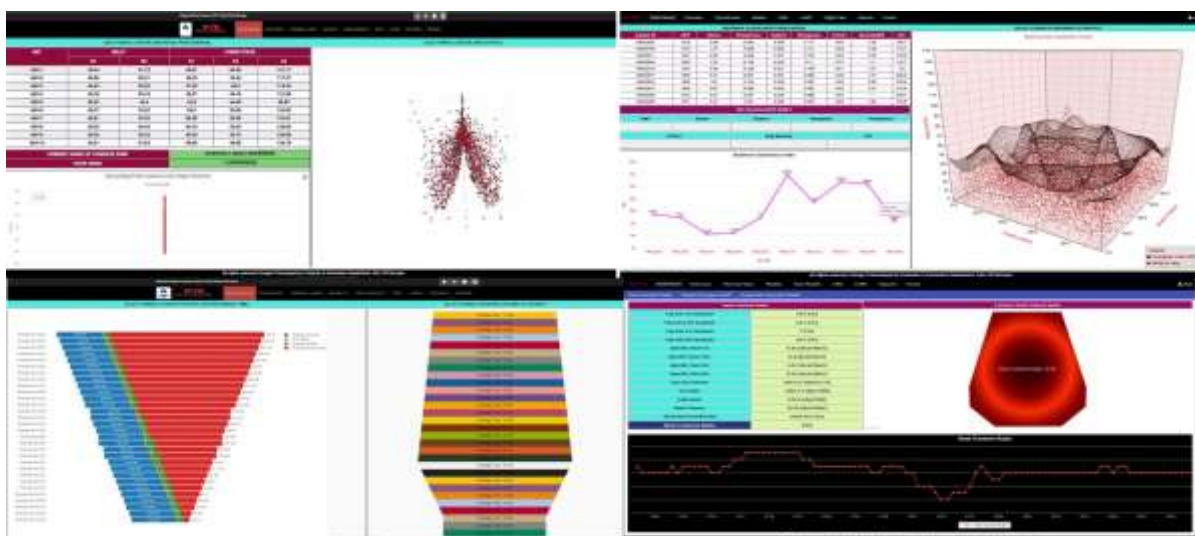
Process Models & Digital Twin

Process Models Developed In-house

Basemix Mass Balance Model	Sinter Quality Prediction Model	Burden Calculation Model	Predictive IGV model for Turboblenders
Burden Distribution Model	Shaft Track Model	Heat Balance Model	Performance Prediction of HP Heaters
Stave Condition Monitoring	Heat Flux Model in BF	Tuyere Health Management	
Cohesive Zone Prediction Model	Hearth Management Model (DCI)	Production Forecast Model in BF	
Operational Index Model in BF	Heat Pacing Model in SMS	Ferro Alloy Consumption Model	

Digital Twin for BF-5

3D Visualization based System having Detailed view of Blast Furnace Process, & Its Equipment in real-time vis-à-vis change in furnace parameters over a period of time.



Digitalization in Safety, Security & Surveillance Systems at ISP

Vehicle Tracking System (VTS)

300 nos. of portable GPS devices have been procured. All road routes of the plant have been mapped. Route taken by vehicle along with information on speed/stops of vehicle is displayed online on real time basis in a large screen in control Room.

Provide 24*7 monitoring of outside vehicles entering the plant for analysis purposes

Pop-up messages for speed violation, undue stoppages and customized Report generation facility is available for deriving useful intelligence inputs.

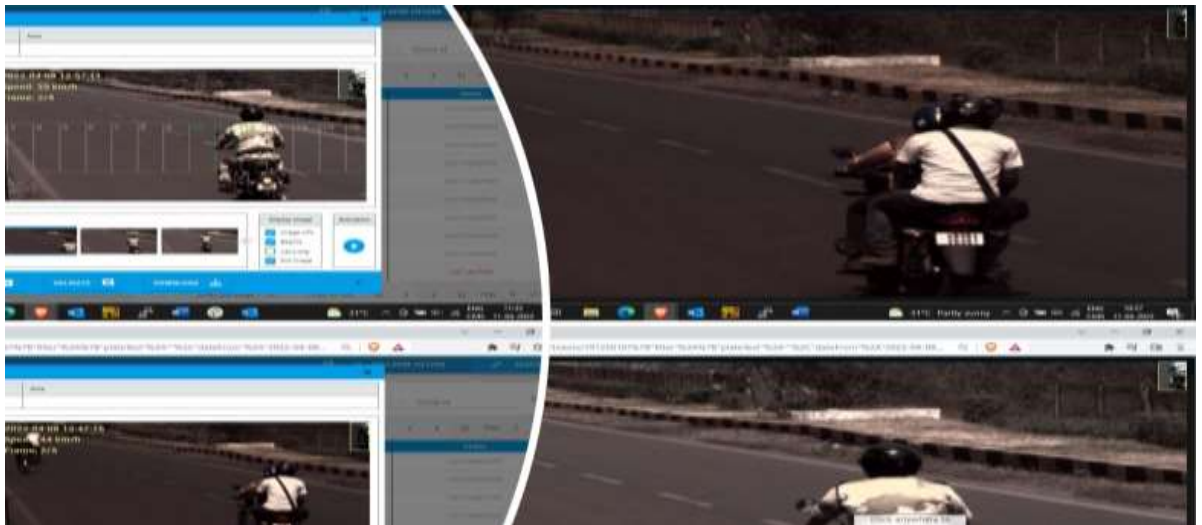
Vehicle Tracking system for improving productivity, efficiency and profitability.

Drone Surveillance at ISP

One no. of Drone has been procured with requisite DGCA clearance. Helps in surveillance of remote and strategic locations during daytime and at night. Handy in gathering information as well as survey of inaccessible locations where CCTV is not installed. Augments existing CCTV based Security system.

Vehicle Speed Monitoring System

Installation of speed Enforcement System and awareness system has been done at 09 locations.



Digitalization in Safety, Security and Surveillance

- **Vehicle Tracking System (VTS):** GPS-enabled tracking of all road vehicles inside the plant. Real-time display of vehicle routes, speed, and stoppages on large screens in control rooms. Automated alerts for speed violations and undue stoppages. Provides intelligence for safety management and productivity improvement.
- **Drone Surveillance:** Drones with DGCA clearance provide surveillance of remote, strategic, and inaccessible locations during day and night. Complement existing CCTV systems and support survey of areas where fixed cameras cannot be installed.
- **Vehicle Speed Monitoring System:** Speed enforcement and awareness systems installed at multiple plant entry/exit points to reduce speeding incidents and improve road safety within the plant.
- **CCTV and AI-based Video Analytics:** Modern surveillance systems incorporate AI-based video analytics for automatic detection of safety violations (PPE non-compliance, unauthorised entry, fire/smoke detection), augmenting human monitoring capacity.